

FinTrust Digital Bank — Risk Intelligence & Predictive Modeling

Track: Data Science

Phase: Week 1 — Problem Definition, Target Assessment & Data Profiling

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1. Executive Context & Business Understanding

FinTrust Digital Bank is scaling its digital financial services across transactions and customer operations. To maintain platform integrity and mitigate operational risk, the bank requires an automated predictive capability to identify high-risk financial transactions in real time.

1.1 Predictive Problem Statement (Part A)

- **Objective:** Predict whether an incoming transaction requires formal risk review (`Risk_Review_Flag` = Yes);
- **Business Value:** Enables proactive risk management, reduces operational workloads for compliance officers, and prevents fraudulent activity before settlement;
- **Primary Stakeholders:** Fraud Operations, Risk & Compliance Team, Customer Support;
- **Success Criteria:** High Precision-Recall AUC (PR-AUC), high F1-Score on the minority risk class, and real-time prediction latency (< 100ms).

1.2 Target Variable Assessment (Part B)

- **Target Field:** `Risk_Review_Flag`
 - **Type & Values:** Binary Categorical (`Yes` / `No`)
 - **Nature:** Synthetic indicator created for risk assessment exercises
 - **Expected Distribution:** Heavily imbalanced (the vast majority of daily banking transactions are legitimate)
 - **Target Limitations:** As a synthetic label, it serves as a proxy for risk review rather than confirmed historical fraud ground truth.
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Customer Dataset Shape: (1500, 12)

Transaction Dataset Shape: (12000, 11)

=== Target Distribution (`Risk_Review_Flag`) ===

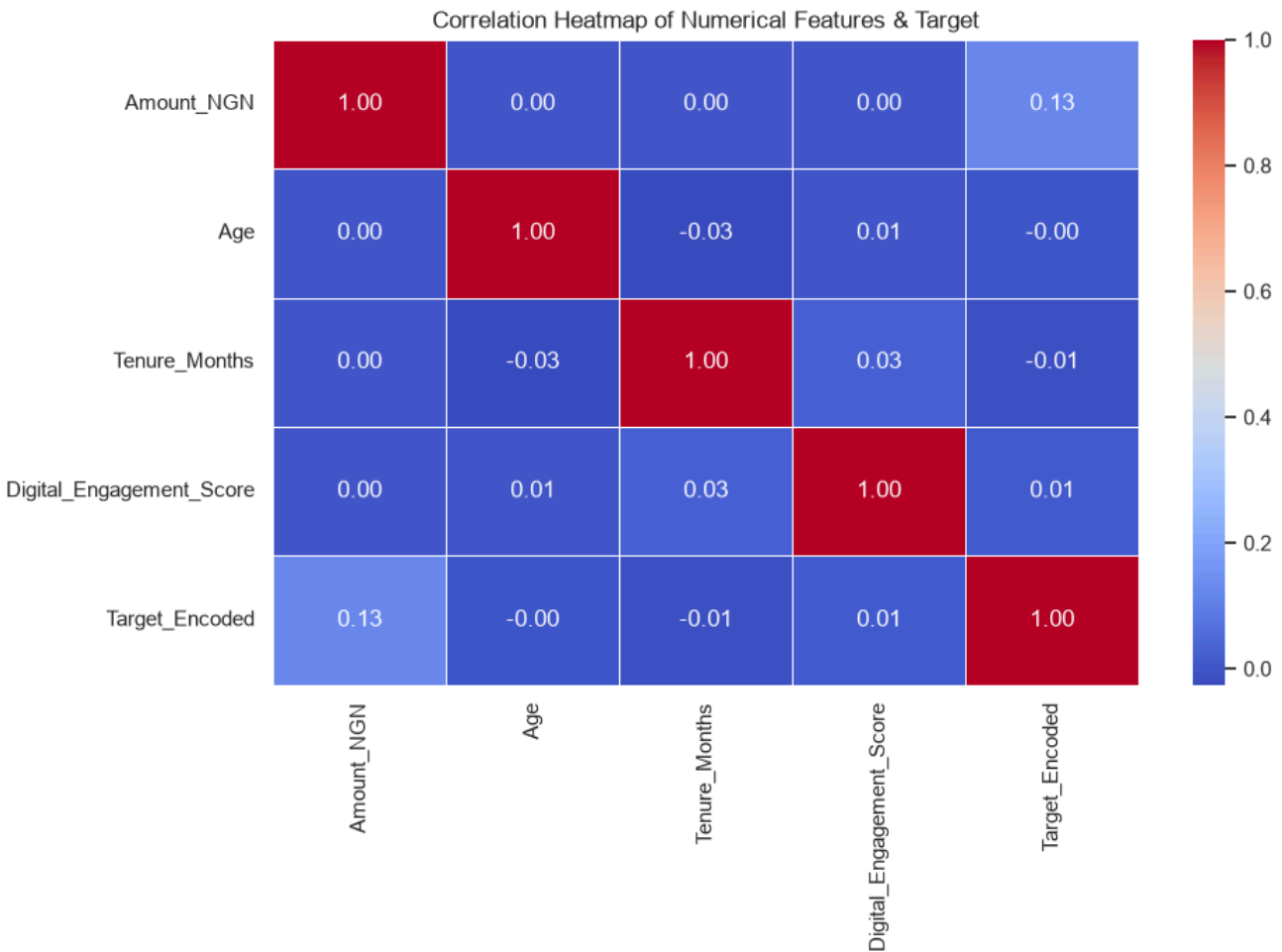
	Count	Percentage (%)
Risk_Review_Flag		
No	9648	80.4
Yes	2352	19.6

1.3. Candidate Features Assessment (Part C)

To prepare the dataset for predictive modeling, we merge the `Customer` and `Transaction` tables using an `inner join` on `Customer_ID`. This provides a consolidated feature space combining demographic attributes with transactional behavior.

Below, we engineer and inspect candidate features across both tables before performing feature selection and dimensionality reduction (e.g., PCA or correlation filtering).

Merged Dataset Shape: (12000, 22)
Columns in Merged Dataset: ['Transaction_ID', 'Customer_ID', 'Transaction_DateTime', 'Transaction_Type', 'Amount_NGN', 'Channel', 'Device_Type', 'Location', 'International_Transaction', 'Transaction_Status', 'Risk_Review_Flag', 'Customer_Name', 'Age', 'Gender', 'City', 'Customer_Segment', 'Account_Type', 'Tenure_Months', 'Digital_Engagement_Score', 'Monthly_Income_Band', 'Preferred_Channel', 'Account_Status']



```
Missing values in merged dataset:
Device_Type      96
Location         96
dtype: int64
```

1.3.1 Correlation Heatmap Interpretation & Feature Selection Methodology

Heatmap Analysis & Limitations

The Pearson correlation matrix above evaluates only **linear relationships between continuous numerical features**:

- **Weak Linear Signal:** Features such as `Age` ($r = -0.00$), `Tenure_Months` ($r = -0.01$), and `Digital_Engagement_Score` ($r = 0.01$) show near-zero linear correlation with `Target_Encoded ( Risk_Review_Flag )`
- **Primary Numerical Signal:** `Amount_NGN` shows a positive linear correlation ($r = 0.13$) with the target, indicating that higher transaction values moderately increase the likelihood of a risk review trigger
- **Methodological Limitation:** Because FinTrust's datasets heavily rely on categorical attributes (`Transaction_Type` , `Channel` , `Device_Type` , `Location` , `International_Transaction` , `Monthly_Income_Band`), linear Pearson correlation is insufficient to determine feature relevance.

Candidate Selection Framework

Instead of relying solely on linear correlation or intuition, feature selection for mixed tabular data relies on a three-tier framework:

1. **Domain Risk Relevance:** Financial risk patterns inherent to transaction channels and user behaviors.
2. **Categorical Association Metrics:** Statistical tests (χ^2 test of independence, Cramér's V, and Mutual Information Score) designed to detect non-linear dependencies between categorical predictors and binary targets.
3. **Behavioral Anomaly Rationale:** Cross-referencing customer baseline profiles with individual transaction occurrences.

1.3.2 Candidate Features Table (Part C Deliverable)

Below is the structured assessment of **12 candidate features** selected across the merged Customer-Transaction dataset:

Feature Name	Source Table	Data Type	Why It May Matter (Risk Rationale)
<code>Amount_NGN</code>	Transaction	Decimal	Primary indicator of financial exposure; high-value transfers trigger automated compliance rules

Feature Name	Source Table	Data Type	Why It May Matter (Risk Rationale)
Transaction_Type	Transaction	Categorical	Specific operations (e.g., Wire Transfers, International Payments) carry higher inherent fraud velocity than domestic merchant payments
Channel	Transaction	Categorical	Risk exposure varies across channels (e.g., Web API vs. Mobile App vs. POS)
Device_Type	Transaction	Categorical	Unrecognized, obsolete, or rapidly switching devices often indicate account takeover
Location	Transaction	Categorical	Geographic anomalies (e.g., transactions originating from unusual cities or high-risk jurisdictions)
International_Transaction	Transaction	Binary Categorical	Cross-border payments carry significantly higher regulatory and fraud risk profiles
Transaction_Status	Transaction	Categorical	Repeated failed or pending transactions prior to execution may signal brute-force or unauthorized attempts
Monthly_Income_Band	Customer	Categorical	Serves as a baseline to evaluate whether Amount_NGN is proportional to the customer's expected financial capacity
Digital_Engagement_Score	Customer	Decimal	Sudden high-value digital transactions from customers with historically low engagement scores

Feature Name	Source Table	Data Type	Why It May Matter (Risk Rationale)
			signal anomalous account usage
Tenure_Months	Customer	Integer	Newly created accounts ("low tenure") statistically present higher early-life fraud and chargeback risk
City	Customer	Categorical	Customer registered residence; useful for engineering a Location_Mismatch feature against transaction Location .
Account_Type	Customer	Categorical	Business vs. Savings accounts have distinct transaction velocity and volume expectations

2. Prepare

2.1 Risk Factors Hypotheses (Part D)

To guide feature engineering and exploratory data analysis, we establish the following preliminary hypotheses regarding factors associated with Risk_Review_Flag :

Methodological Note: In accordance with project guidelines, these statements represent provisional research hypotheses to be tested statistically, rather than definitive empirical conclusions.

Hypothesis 1: Transaction Value relative to Customer Income

- **Research Hypothesis:** Transactions where the monetary amount (Amount_NGN) is disproportionately high relative to the customer's Monthly_Income_Band are more likely to be flagged for risk review.
- **Null Hypothesis (H_0):** There is no significant difference in the amount-to-income ratio between transactions flagged for risk review (Risk_Review_Flag = Yes) and unflagged transactions (Risk_Review_Flag = No).
- **Alternative Hypothesis (H_1):** Transactions flagged for risk review exhibit a significantly higher amount-to-income ratio than unflagged transactions.

- **Planned Statistical Test:** Non-parametric Mann-Whitney U Test / Two-sample t-test.
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Hypothesis 2: Device Type Verification and Anomaly

- **Research Hypothesis:** Transactions originating from missing or unverified device types (`Device_Type`) carry a higher probability of being flagged for risk review compared to recognized devices.
 - **Null Hypothesis (H_0):** The presence of a missing/unverified `Device_Type` is statistically independent of the `Risk_Review_Flag` status.
 - **Alternative Hypothesis (H_1):** Transactions with missing/unverified `Device_Type` categories are significantly associated with a higher proportion of risk review flags.
 - **Planned Statistical Test:** Chi-Square (χ^2) Test of Independence.
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Hypothesis 3: Customer Tenure and Account Maturity

- **Research Hypothesis:** Newer accounts (low `Tenure_Months`) executing high-frequency transfers present a higher likelihood of risk review flags than established accounts.
 - **Null Hypothesis (H_0):** Customer tenure length has no relationship with the transaction risk review rate.
 - **Alternative Hypothesis (H_1):** Customers with lower account tenure have a statistically higher risk review flag rate during transfer operations.
 - **Planned Statistical Test:** Logistic Regression odds ratio evaluation / Point-biserial correlation test.
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Hypothesis 4: Cross-Border & Channel Exposure

- **Research Hypothesis:** Cross-border transactions (`International_Transaction = Yes`) performed via web channels exhibit a higher risk review rate than domestic mobile app transactions.
 - **Null Hypothesis (H_0):** International transaction status and channel selection do not affect the distribution of the `Risk_Review_Flag` .
 - **Alternative Hypothesis (H_1):** International transactions conducted via web channels have a significantly higher probability of being flagged for risk review.
 - **Planned Statistical Test:** Two-way ANOVA / Chi-Square (χ^2) Test on interaction terms.
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2.2. Initial Modelling Plan (Part E)

The proposed end-to-end Machine Learning plan for the FinTrust Risk Review problem covers 9 steps:

2.2.1 Data Preparation

- Merge `Customer` and `Transaction` datasets on `Customer_ID`
- Handle missing values in `Device_Type` and `Location` using explicit "Unknown" category encoding or mode imputation
- Encode categorical variables using One-Hot Encoding for nominal variables and Ordinal Encoding for `Monthly_Income_Band`

- Apply `RobustScaler` or `StandardScaler` to numerical attributes (`Amount_NGN` , `Tenure_Months` , `Digital_Engagement_Score`) to handle skewness.

2.2.2 Exploratory Analysis (EDA)

- Analyze the target distribution of `Risk_Review_Flag` to quantify class imbalance
- **Testing Hypothesis 1 (H_1):** Perform a Mann-Whitney U test comparing `Amount_NGN` distributions across `Risk_Review_Flag` classes
- **Testing Hypothesis 2 (H_2):** Execute a Chi-Square (χ^2) Test of Independence between missing/unverified `Device_Type` and `Risk_Review_Flag`
- **Testing Hypothesis 3 (H_3):** Measure correlation and odds ratios between `Tenure_Months` and transfer risk flags
- **Testing Hypothesis 4 (H_4):** Analyze interaction effects between `International_Transaction = Yes` , `Channel` , and the risk target

2.2.3 Feature Engineering

- **Relative Transaction Ratio:** Ratio of `Amount_NGN` to the estimated midpoint of `Monthly_Income_Band`
- **Device Anomaly Flag:** Binary indicator representing whether `Device_Type` is missing or unverified
- **Location Mismatch Flag:** Comparison between customer registered `City` and transaction `Location`
- **Dimensionality Reduction:** Dimensionality reduction via Principal Component Analysis (PCA) on continuous behavioral features to extract uncorrelated variance components. If necessary

2.2.4 Baseline Model

- Implement a simple **Logistic Regression** and a **DummyClassifier** (predicting majority class/stratified) to establish lower-bound performance benchmarks.

2.2.5 Candidate Algorithms

- **Supervised Classifiers:** Random Forest Classifier, XGBoost / LightGBM, and a PyTorch Multi-Layer Perceptron (MLP) for non-linear risk scoring
- **Unsupervised Anomaly Detection:** K-Means Clustering and Isolation Forest / Autoencoders trained exclusively on normal transactions (`Risk_Review_Flag = No`) to generate anomaly risk scores
- **Hybrid Ensemble:** Combining supervised probability scores with unsupervised anomaly distance metrics to produce a unified Risk Score

2.2.6 Evaluation Metrics

- Standard accuracy is strictly excluded due to severe class imbalance
- Primary evaluation metrics: **Precision-Recall Curve (PR-AUC)**, **F1-Score** (focused on the positive `Risk_Review_Flag = Yes` class), and **Recall** (to minimize false negatives in fraud screening).

2.2.7 Validation Strategy

- Utilize **Stratified 5-Fold Cross-Validation** to maintain class ratio integrity across folds

- If temporal patterns exist in `Transaction_DateTime`, evaluate a time-based train/test split to prevent data leakage

2.2.8 Error Analysis

- Construct Confusion Matrices to analyze False Positives (legitimate transactions blocked) vs False Negatives (risky transactions missed)
- Extract Feature Importances (Gini / SHAP values) to verify whether features derived from hypotheses H_1 through H_4 serve as primary decision drivers for the trained models

2.2.9 Model Limitations

- `Risk_Review_Flag` is a synthetic educational label and may not perfectly reflect real-world bank fraud mechanisms
- Static batch modeling lacks immediate adaptation to novel fraud tactics without periodic retrain pipelines.